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Help more people find respirators that fit

Extend MasqFit to Android and standard cameras, and test facial features tailored to different respirator styles.

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Draft for discussion | Android and web research | Open research outputs

> **Proposed support: $196,800, with an initial $66,800 option.**
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> The full request includes $180,000 for research and delivery plus a separately proposed $16,800 parental-leave allowance. The initial option funds $50,000 of feasibility work and reserve plus the leave allowance; continuation of $130,000 is conditional on review. Budget and start date are negotiable.

The opportunity

MasqFit is an open-source iOS app that uses facial measurements and contributed fit-test results to recommend respirators. Kanro's original $50,000 grant supported its development. Its current scan-based workflow depends on compatible iPhone hardware. Ordinary RGB cameras could extend access to Android and web users while producing reusable evidence about low-cost facial measurement.

The proposed public-health benefit is a more informed shortlist of respirator models and sizes to test. Recommendations complement formal fit testing. The research will measure selection usefulness rather than infer protection from geometric accuracy alone.

See the existing iPhone capture → measurements → mask-ranking flow in three screenshots →

> **Potential global smartphone reach: roughly 3–4× an iPhone-only audience.**
>
> Supporting Android and ordinary RGB cameras could remove the need for compatible iPhone depth hardware. October 2026 mobile web-usage shares imply a combined Android/iOS base about 3.62× the iOS-only base; active-device estimates suggest an approximately fourfold global smartphone base relative to iPhones. These are access proxies, not unique-user counts or adoption forecasts. Browser capture could also serve supported webcams and iPhones without the current depth-hardware requirement.
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> Statcounter, October 2026 · Counterpoint, 2025 active installed base. Actual reach depends on validated hardware, capture usability, processing access, and respirator availability.

Recent experimental progress

A working experimental pipeline now reconstructs facial surfaces from multiple ordinary-camera photos, applies Gaussian smoothing by 3D convolution, and sets physical scale using a tape-measured bitragion–subnasale arc: from one ear's tragion, beneath the nose, to the other tragion.

0.46%	1.37%	1.98%
Nose-width absolute relative error	Face-width absolute relative error	Face-length absolute relative error

Exploratory comparison from one source capture sequence, scaled with a separate tape arc. Endpoint review is pending; these are not population accuracy estimates.

The corresponding estimates are 35.16 versus 35.00 mm for nose width, 152.97 versus 150.90 mm for face width, and 112.33 versus 114.60 mm for face length. The calibration arc was not refitted to those measurements. These promising comparisons follow iterative method development on the same capture; anatomical review, repeat captures, and independent participants are still needed.

Existing iOS recommender: top-1 and top-3 results

The recommender used with MasqFit's iOS workflow reports a **51.7% top-1 hit rate** and an **82.2% top-3 hit rate** in retrospective evaluation. These are unweighted averages across five user-grouped cross-validation folds, among held-out users with at least one positive fit label.

Example: if a person's previously tested observations rank as negative, negative, positive, then top-1 misses and top-3 succeeds. Top-3 means at least one positive result appears within the first three ranked observations; it does not require all three to fit.

Live dashboard checked November 9, 2026 · Model November 8, 2026. The candidate set consists of recorded test observations, with mixed-source labels and possible repeats, rather than three new masks from the full catalog. Prospective success for new users remains to be measured.

Illustrative example · not additional study data

1 · Turn one person's ranking into two hit scores

Example B has at least one positive held-out fit label, so this person is eligible. The model ranks their previously tested observations from highest to lowest predicted score:

RANK 1	RANK 2	RANK 3	RANK 4
Observation A	**Observation B**	**Observation C**	**Observation D**
Negative label	Negative label	Positive label	Negative label

Top 3 includes one positive result

Top-1: miss → 0

Only rank 1 counts. Its label is negative.

Top-3: hit → 1

Ranks 1–3 count. At least one label is positive.

The top-3 score is 1, not 1/3: this asks whether the shortlist contains any positive result.

Person → 0 or 1

Score a hit for each eligible held-out person.

Within each fold

Divide hit people by eligible people, separately for top-1 and top-3.

Across five folds

Add the five fold percentages and divide by 5. Each fold gets equal weight.

Train on four person groups; evaluate on the fifth, repeating until every group has been held out. All observations from a person stay together. The reported means are 51.7% top-1 and 82.2% top-3; the actual fold counts are not available in the dashboard snapshot. See a worked five-fold example →

Improve selection with features tailored to respirator style

Alongside broader access, test whether different facial features improve recommendations for **filtering facepiece respirators (FFRs) and quarter-mask elastomerics**. The existing iOS recommender already uses mask family/style terms, while its facial geometry is summarized by a perimeter proxy. The new work would retain local geometry separately: for example, cheek/chin dimensions for flexible lower panels, nasal slope and bridge shape for the upper seal, and dimensions matched to a quarter-mask's actual contact footprint.

These are hypotheses to test on feature-linked human fit outcomes. Compare style-dependent features with the current model using the same held-out people and candidate observations; preserve the baseline where added features do not help. Quarter-mask elastomeric work begins as a bounded feasibility analysis, with a larger trial dependent on adequate data and separately agreed scope.

See both experimental paths on the 3D model →

Which approach delivers the most useful result for its cost?

The research will compare **photogrammetry with 3D smoothing, direct-photo-to-FLAME fitting, and the existing iPhone-only ARKit workflow** on the same people. The working hypothesis is a tradeoff between local-shape accuracy and speed, capture effort, and processing cost—not a demonstrated ordering of the three methods.

For healthcare personnel onboarding or respirator-selection campaigns, measure time to a usable recommendation, device/compute and staff costs, retries, and quantitative fit-test outcomes. Ask how much more photogrammetry costs than direct fitting, and whether any extra accuracy changes the shortlist or produces a material increase in first-choice fit-test passes. A small geometric improvement with unchanged decisions may offer little operational value.

No comparable per-user cost or head-to-head accuracy gain has yet been established. The study will report cost per usable recommendation and per verified first-choice fit-test pass, with uncertainty. A numerical example and the comparison protocol appear in the detailed proposal.

What funding would establish

1. **Feasibility and calibration:** freeze landmark definitions and capture settings, test smoothing and scale sensitivity, and complete a 20-adult geometry pilot.
2. **Independent measurement validation:** expand to 120 adults total, with 60 development participants including the pilot and 60 untouched test participants. Compare repeated captures with independently acquired physical/reference measurements, across devices and face shapes.
3. **Selection usefulness:** contingent on measurement performance, enroll 60 separate adults in a prospective quantitative fit-testing pilot, comparing the first frozen camera-based recommendation with each participant's initial choice. Release an Android/web research beta within validated conditions.

Deliverables include reproducible measurement code, a documented benchmark and failure analysis, device/usability results, a prospective fit-pilot report, and public methods and non-identifying aggregate findings. Participant images and identifiable meshes require separate consent for sharing.

Duration, milestones, and partnership

The full project is planned for 12 active research months, approximately 14 calendar months if it spans eight weeks of parental leave. The initial phase covers three active months. Start dates will be agreed with Blueprint given its evolving grant workflow. If funding begins after leave, we will rebase the timeline and allowance rather than assume retroactive coverage.

Breathesafe LLC would agree public-benefit, licensing, and dissemination terms with Blueprint before award.

Requested next step: feedback from Victoria and Nickia on program fit, the initial phase, and the evidence required for a funding decision.

Read the detailed proposal →

Prepared following Blueprint's invitation to review a concept note. Review does not imply a funding commitment or a decision timeline. PPE program context · NIOSH fit-testing guidance.